

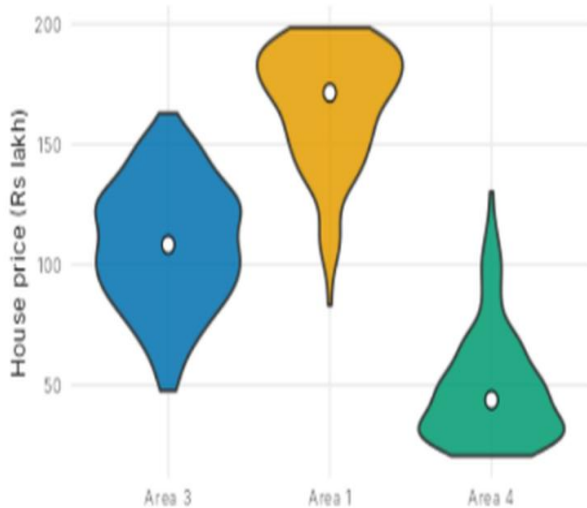
R.No: 6

Reg. No: 192324244

Name: B KIRTHANAA

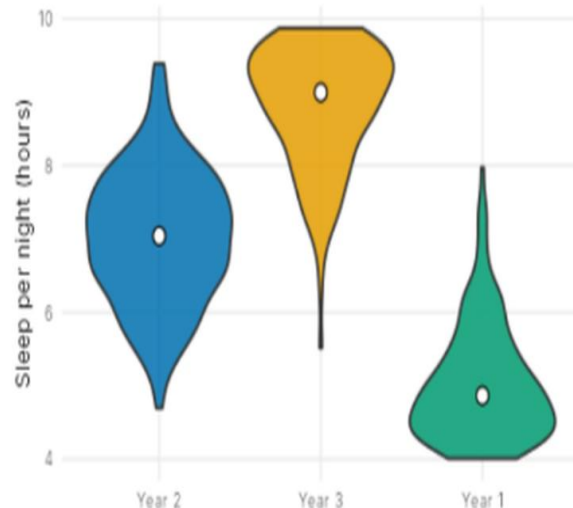
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: House price (Rs lakh)



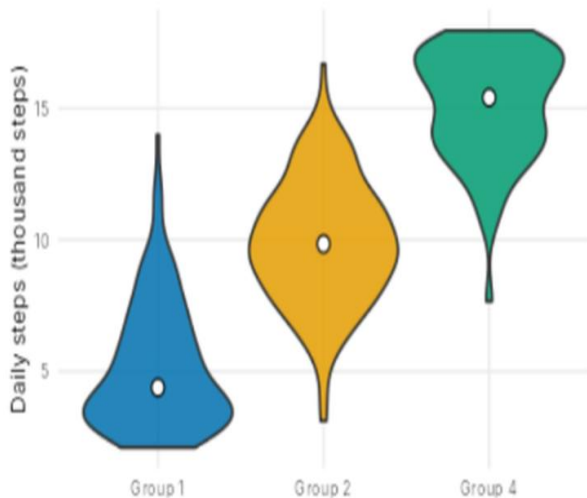
Violin	Skew (circle one)	Mean vs Median (circle one)
Area 3	Right / Left / Symm.	Mean > / < / ~ Median
Area 1	Right / Left / Symm.	Mean > / < / ~ Median
Area 4	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Sleep per night (hours)



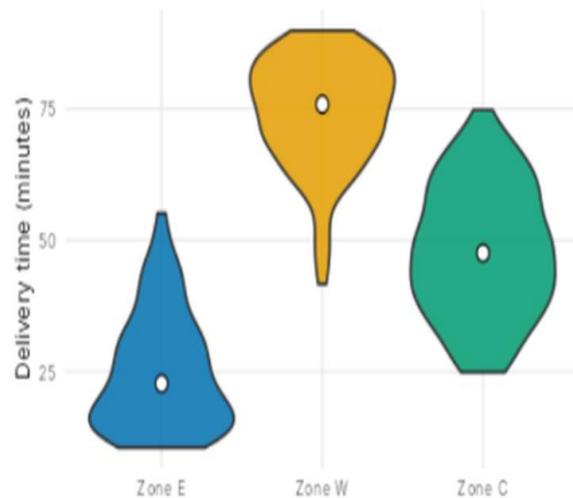
Violin	Skew (circle one)	Mean vs Median (circle one)
Year 2	Right / Left / Symm.	Mean > / < / ~ Median
Year 3	Right / Left / Symm.	Mean > / < / ~ Median
Year 1	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Daily steps (thousand steps)



Violin	Skew (circle one)	Mean vs Median (circle one)
Group 1	Right / Left / Symm.	Mean > / < / ~ Median
Group 2	Right / Left / Symm.	Mean > / < / ~ Median
Group 4	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Delivery time (minutes)



Violin	Skew (circle one)	Mean vs Median (circle one)
Zone E	Right / Left / Symm.	Mean > / < / ~ Median
Zone W	Right / Left / Symm.	Mean > / < / ~ Median
Zone C	Right / Left / Symm.	Mean > / < / ~ Median

How to read a violin plot

- The width of the violin at any height shows how many values lie there (the density). The white dot is the median.
- Wide at the bottom with a long thin tail going up: Right skew, and the mean is pulled above the median (Mean > Median).
- Wide at the top with a long thin tail going down: Left skew, and the mean is pulled below the median (Mean < Median).
- Balanced, mirror-like shape on both sides of the median dot: Symmetric, and the mean is about equal to the median (Mean ~ Median).

PART A: REASONING FOR EACH VIOLIN:

Violin	Skew	Reason	Mean vs Median
Area 3	Symmetric	Balanced shape around the median dot (about 108 lakh); no long tail on either side.	Mean ~ Median
Area 1	Left	Most houses are at high prices (median about 170 lakh); a thin tail stretches down to low prices.	Mean < Median
Area 4	Right	Most houses are at low prices (median about 45 lakh); a long thin tail stretches up to about 130 lakh.	Mean > Median
Year 2	Symmetric	Evenly spread around the median of about 7 hours.	Mean ~ Median
Year 3	Left	Most students sleep long hours (median about 9); a tail extends down towards short sleep.	Mean < Median
Year 1	Right	Most students sleep few hours (median about 4.8); a tail extends up to about 8 hours.	Mean > Median
Group 1	Right	Bulk of the data is at low step counts (median about 4.5 thousand); a long tail goes up to about 14.	Mean > Median
Group 2	Symmetric	Balanced shape around the median of about 10 thousand steps.	Mean ~ Median
Group 4	Left	Bulk of the data is at high step counts (median about 15 thousand); a tail goes down towards 8.	Mean < Median
Zone E	Right	Most deliveries are quick (median about 23 min); a tail goes up to slow deliveries.	Mean > Median
Zone W	Left	Most deliveries take long (median about 75 min); a tail goes down to faster deliveries.	Mean < Median
Zone C	Symmetric	Balanced shape around the median of about 47 min.	Mean ~ Median

PART B:

B1. In Chart 2, violin "Year 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical sleep per night here: mean or median? Give one reason. [1]

Ans: The tail of "Year 3" is towards the left (the lower values / fewer hours of sleep), so it is left-skewed. The median is the better measure of a typical sleep per night, because the few very low values in the tail pull the mean downward, while the median is not affected by these extreme values and stays near where most of the data lies.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Ans: `ggplot(df, aes(x = group, y = value, fill = group)) + geom_violin() + stat_summary(fun = median, geom = "point", colour = "white", size = 3)`

Explanation of the B2 code

- `ggplot(df, aes(x = group, y = value, fill = group))`: takes the data frame `df`, puts `group` on the x-axis, `value` on the y-axis and colours each violin by `group`.
- `geom_violin()`: draws one violin (a mirrored density curve) for each `group`.
- `stat_summary(fun = median, geom = "point", colour = "white", size = 3)`: calculates the median of each `group` and shows it as a white dot, as in the charts.

Summary

Chart	Left skew (Mean < Median)	Symmetric (Mean ~ Median)	Right skew (Mean > Median)
Chart 1: House price (Rs lakh)	Area 1	Area 3	Area 4
Chart 2: Sleep per night (hours)	Year 3	Year 2	Year 1
Chart 3: Daily steps (thousand steps)	Group 4	Group 2	Group 1
Chart 4: Delivery time (minutes)	Zone W	Zone C	Zone E

Conclusion

Violin plots show the full shape of a distribution, not just a single summary number. A long tail pulls the mean towards it, so in skewed data the median is the more reliable measure of a typical value, while in symmetric data the mean and median are almost the same.